

CONTACT

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RESEARCH INTEREST

State Estimation

Kalman filters
Graph Optimization
GNSS & SLAM & Sensor Fusion

Variational Inference

Variational Bayes
Parameter Tuning
Online Model Identification

Deep/Reinforcement Learning

Object Recognition
Active Perception

HAOMING ZHANG

Robotics - Mechanical Engineering

EDUCATION

Ph. D. - Robotics
RWTH Aachen University, Germany

2020 - 2025

M. Sc. - Mechatronics
University of Duisburg-Essen, Germany

2015 - 2016

B. Sc. - Mechanical Engineering
University of Duisburg-Essen, Germany

2012 - 2015

WORK EXPERIENCE

Postdoctoral Research Associate
Learning Systems and Robotics Lab, TUM

Research on state estimation and learning-based inference;
Project: autonomous drone

Oct 01 - ongoing

Research Associate
Institute of Automatic Control, RWTH Aachen Uni.

Research on state estimation and learning-based inference;
Project: autonomous shipping

Jul 20 - Aug. 24

Research Associate
Cybernetics Lab, RWTH Aachen Uni.

Research/Teaching: state estimation and trajectory planning;
Project: human/robot collaboration and mobile manipulation

May 17 - Jun 20

Student Research Associate
Chair of Dynamics and Control, Uni. Duisburg-Essen

Research: simulation design for autonomous driving

May 16 - Feb 17

PUBLICATIONS

◊: first / equally-contributed author

SVN-ICP: Uncertainty Estimation of ICP-based LiDAR Odometry using Stein Variational Newton Method
IEEE Robotics and Automation Letters

DOI: 10.1109/LRA.2025.3619836

◊ 2025

Handling GNSS-Outliers in Multisensor State Estimation
Dissertation

◊ 2025

Robust Statistics vs. Machine Learning vs. Bayesian Inference: Insights into Handling Faulty GNSS Measurements in Field Robotics
Workshop SIAV-FM2L at IROS2025

DOI: 10.48550/arXiv.2503.04933

◊ 2025

SKILLS

C++ & Python	7+ yrs
ROS	7+ yrs
Machine Learning	5+ yrs
Teaching	3+ yrs
Web Development	<1 yrs

LANGUAGES

English
Proficient (C1)
German
Fluent (C2)
Chinese
Native

Learning-based GNSS Uncertainty Quantification using Continuous-Time Factor Graph Optimization 1st German Robotics Conference DOI: 10.48550/arXiv.2504.06015	◇ 2025
GNSS/Multi-Sensor Fusion Using Continuous-Time Factor Graph Optimization for Robust Localization IEEE Transactions on Robotics DOI: 10.1109/TRO.2024.3443699	◇ 2024
Learning-based NLOS Detection and Uncertainty Prediction of GNSS Observations with Transformer-Enhanced LSTM Network Proc. of IEEE 26th Int. Conf. on Intelli. Trans. Syst. DOI: 10.1109/ITSC57777.2023.10422672	◇ 2023
Object Detection and Heading Estimation from Radar Raw data Proc. of 34th IEEE Intelligent Vehicles Symposium DOI: 10.1109/IV55152.2023.10186591	2022
Continuous-Time Factor Graph Optimization for Trajectory Smoothness of GNSS/INS Navigation in Temporarily GNSS-Denied Environments IEEE Robotics and Automation Letters DOI: 10.1109/LRA.2022.3189824	◇ 2022
Predicting basin stability of power grids using graph neural networks New Journal of Physics DOI: 10.1088/1367-2630/ac54c9	2022

PROJECTS

MuSEAS: multisensor dataset with annotations for ship automation Role: Project Leader Visual odometry; Multi-sensor fusion; Mapping; Hardware/Software concept; Project management	2023-2024
FernBin: Remote-controlled and coordinated driving in land shipping Role: Project Leader Robust ship localization; Sensor fusion; Hardware/Software concept; Project management	2020-2024
UrbANT: urban, automated, user-oriented transport platform Role: Project Collaborator Outdoor Robot localization; Object detection/tracking; Robot design and prototyping; Project management	2019-2020
ARIZ: work in the industry of the future Role: Project Collaborator Indoor robot localization/SLAM; Trajectory planning of mobile/stationary robots	2017-2019

ACTIVITIES

4th Workshop on Intelligent Vehicle Meets Urban: Safe and Certifiable Navigation and Control for Intelligent Vehicles in Complex Urban Scenarios
Role: Co-Organizer

Gold Coast, AU

[More Information](#)

3rd Workshop on Intelligent Vehicle Meets Urban: Safe and Certifiable Navigation and Control for Intelligent Vehicles in Complex Urban Scenarios
Role: Co-Organizer and Moderator

Edmonton, CA

[More Information](#)

2nd Workshop on Intelligent Vehicle Meets Urban: Safe and Certifiable Navigation and Control for Intelligent Vehicles in Complex Urban Scenarios
Role: Co-Organizer and Moderator

Bilbao, ES

[More Information](#)

Demonstrations of 33rd IEEE Intelligent Vehicles Symposium
Role: Organizer

Aachen, DE

[More Information](#)

Talk: Closing the Estimation Loop: Learning-based Uncertainty Quantification for Robust Vehicle Localization
Role: Presenter

Beijing, CN

Tsinghua University, 2024

Talk: Towards Robust Navigation Solution and Flexible Sensor Fusion in Challenging Inland Shipping Scenarios
Role: Presenter

Duisburg, DE

The Autonomous Inland and Short Sea Shipping Conference, 2022

Reviewer for:
IEEE T-RO/RA-L/T-AES/T-WC/Sensors/JSAC
IEEE ICRA/IROS

AoE

References

AoE

Prof. Dr. Timothy D. Barfoot (tim.barfoot@utoronto.ca)
Prof. Dr. Heike Vallery (h.vallery@irt.rwth-aachen.de)
Prof. Dr. Dirk Abel (d.abel@irt.rwth-aachen.de)
Prof. Dr. Li-Ta Hsu (lt.hsu@polyu.edu.hk)